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Experiment: Crypt Arithmetic Problem

Aim

To solve the Crypt Arithmetic Problem using Python by assigning unique digits to letters such that the arithmetic equation is satisfied.

Objectives

- To understand the Crypt Arithmetic Problem.
- To apply brute-force search to find a valid solution.
- To ensure that each letter is assigned a unique digit.
- To implement the solution using Python.
- To verify the correctness of the arithmetic equation.

Algorithm

1. Identify all unique letters in the crypt arithmetic equation.
2. Generate all possible digit assignments (0–9) for the letters.
3. Ensure that no two letters have the same digit.
4. Check that leading letters are not assigned zero.
5. Substitute the digits into the words.
6. Verify whether the arithmetic equation is satisfied.
7. Display the valid solution.

Python Program

```
from itertools import permutations
```

```
letters = ('S', 'E', 'N', 'D', 'M', 'O', 'R', 'Y')
```

```
for p in permutations(range(10), len(letters)):
```

S, E, N, D, M, O, R, Y = p

Leading letters cannot be zero

if S == 0 or M == 0:

continue

send = 1000*S + 100*E + 10*N + D

more = 1000*M + 100*O + 10*R + E

money = 10000*M + 1000*O + 100*N + 10*E + Y

if send + more == money:

print("Solution Found")

print("SEND =", send)

print("MORE =", more)

print("MONEY =", money)

print("\nLetter Assignments:")

print("S =", S)

print("E =", E)

print("N =", N)

print("D =", D)

print("M =", M)

print("O =", O)

print("R =", R)

print("Y =", Y)

break

Sample Output

Solution Found

SEND = 9567

MORE = 1085

MONEY = 10652

Letter Assignments:

S = 9

E = 5

N = 6

D = 7

M = 1

O = 0

R = 8

Y = 2

Out put

```
5- for p in permutations(range(10), len(letters)):
6     S, E, N, D, M, O, R, Y = p
7
8     # Leading letters cannot be zero
9-     if S == 0 or M == 0:
10         continue
11
12     send = 1000*S + 100*E + 10*N + D
13     more = 1000*M + 100*O + 10*R + E
14     money = 10000*M + 1000*O + 100*N + 10*E + Y
15
16-     if send + more == money:
17         print("Solution Found")
18         print("SEND =", send)
19         print("MORE =", more)
20         print("MONEY =", money)
21         print("\nLetter Assignments:")
22         print("S =", S)
23         print("E =", E)
24         print("N =", N)
25         print("D =", D)
26         print("M =", M)
27         print("O =", O)
28         print("R =", R)
29         print("Y =", Y)
30         break
```

Solution Found

SEND = 9567

MORE = 1085

MONEY = 10652

Letter Assignments:

S = 9

E = 5

N = 6

D = 7

M = 1

O = 0

R = 8

Y = 2

=== Code Execution Successful ===

Result

The Crypt Arithmetic Problem was successfully solved using Python. The program assigned unique digits to each letter and verified that the equation **SEND + MORE = MONEY** was satisfied.

Conclusion

The experiment demonstrated how the **brute-force search** technique can be used to solve a crypt arithmetic problem. By checking all valid digit assignments while ensuring unique values and non-zero leading digits, the program successfully found the correct solution. This experiment helps in understanding constraint satisfaction problems and search techniques used in Artificial Intelligence.